T006 / PHASE 1 — MACHINE LEARNING

Urban Microclimate Modelling

Can simple building-layout summaries predict canopy-averaged wind speed on unseen plan families?

Dataset: UrbanTALES — idealised building arrays (224 CFD cases)

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UrbanTALES data © original authors, CC BY-NC 4.0. A scalar-surrogate evaluation — not a spatial-flow or temperature model.

CFD resolves flow around buildings, but rerunning it for every candidate layout is too slow

The physical question

How do building arrangement and wind direction change the average wind speed inside the urban canopy?

The modelling idea

Learn a fast map from layout-level geometric summaries and wind direction to one canopy-averaged wind speed (m/s).

The boundary

This phase predicts a scalar. It does not predict local wakes, street-canyon patterns, a velocity field or temperature.

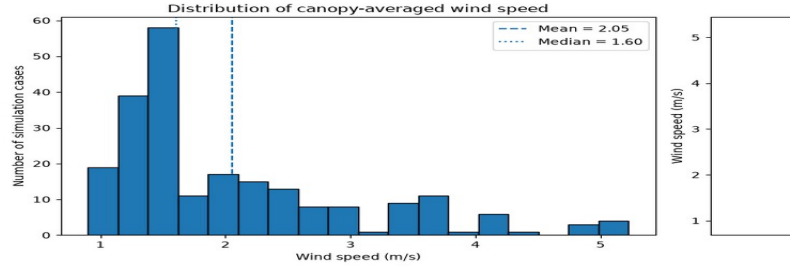
Goal: test whether simple geometry features carry useful signal on plan families the model has never seen.

224 idealised CFD cases: each layout is simulated at 0° and 45° wind

224	112	16	2
simulation cases	unique layouts	plan families	wind directions

What the target looks like

Target: canopy-averaged U (m/s) from ground to mean building height.
Range 0.90–5.22 m/s; mean 2.05, median 1.60 (right-skewed).
Sparse layouts average 3.04 m/s vs 1.37 for dense.
Mean 0° speed 2.24 vs 1.86 at 45°.
All cases share a 16 m mean height, so height is not a free variable.



Distribution of canopy-averaged wind speed across the 224 cases (ML_02).

Small, idealised sample: not a complete picture of real cities.

Models see layout summaries and wind direction only, never the target or any group ID

Plan geometry

Plan-area density and its square capture built-up coverage and a simple nonlinear effect.

Building heights

Std, min, max, range, skew and fraction above the mean describe height variation.

Wind & layout type

Wind direction as sin/cos, aligned/staggered layout and density class. With only 0° and 45°, sin/cos acts as a two-state flag.

Leakage guardrail: target, simulation/layout/family IDs, raw direction and the catalogue label are excluded.

Pipeline: audit raw files (ML_01) → EDA (ML_02) → leakage audit and features (ML_03) → grouped splits (ML_04) → models (ML_05–06) → comparison (ML_07).

Validation holds out whole plan families, so scores reflect genuinely unseen geometry

Fold 1	Fold 2	Fold 3	Fold 4	Fold 5
4 families 56 cases	3 families 42 cases	3 families 42 cases	3 families 42 cases	3 families 42 cases

Primary: family_5fold (GroupKFold by family) — each fold holds out entire families; one fold is the test set in turn.

Secondary: family_loo

Leave one of the 16 families out at a time: the strictest check.

Comparison: layout_5fold

Hold out layouts only. Sibling layouts in training make this an easier, interpolation-style test.

Preprocessing is fitted on training folds only; train/test families and layouts are checked for overlap.

Four models of increasing flexibility, all with fixed settings

Mean baseline

Predicts the training-fold mean. The no-feature reference.

Ridge regression

Linear with an L2 penalty and standardised inputs. Transparent baseline.

Random Forest

Tree ensemble for nonlinear patterns and interactions.

XGBoost

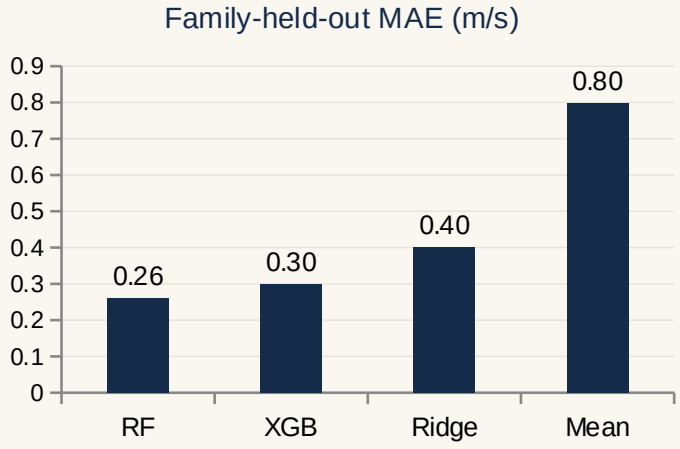
Gradient-boosted trees: does boosting lower the error?

Metrics: MAE (m/s), RMSE, normalised RMSE (÷ target range) and R^2 vs the mean-only reference.

Random Forest cuts error by 67% versus the mean baseline: about 0.26 m/s, ~13% of mean wind speed

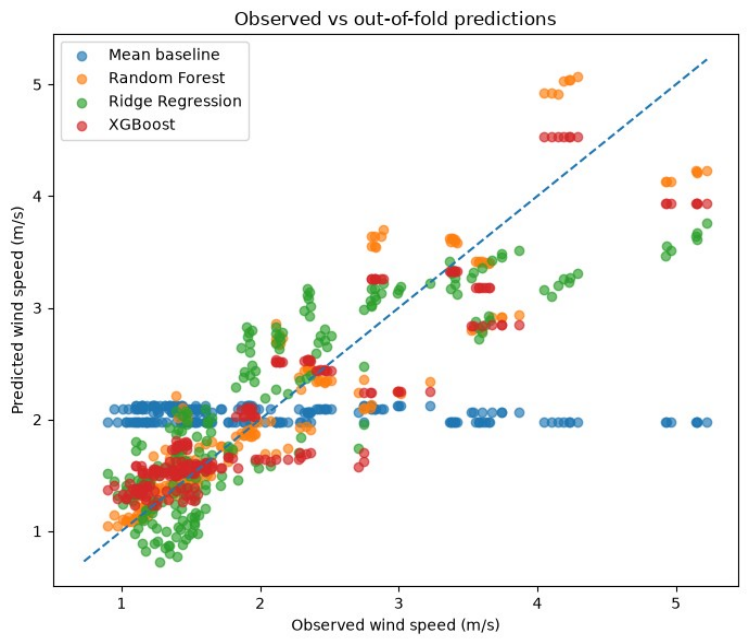
Model	MAE (m/s)	RMSE (m/s)	NRMSE	R^2
Random Forest	**0.259**	**0.384**	**0.089**	**0.851**
XGBoost	0.298	0.401	0.093	0.838
Ridge regression	0.400	0.507	0.117	0.740
Mean baseline	0.798	1.011	0.234	−0.033

Pooled out-of-fold predictions, family_5fold, engineered features, 224 cases / 16 families. Ridge also clearly beats the baseline, so geometry carries real signal.



Takeaway: simple geometry summaries predict canopy wind on unseen families; nonlinear trees help.

Predictions track observations well; the largest misses are at the fast, sparse end



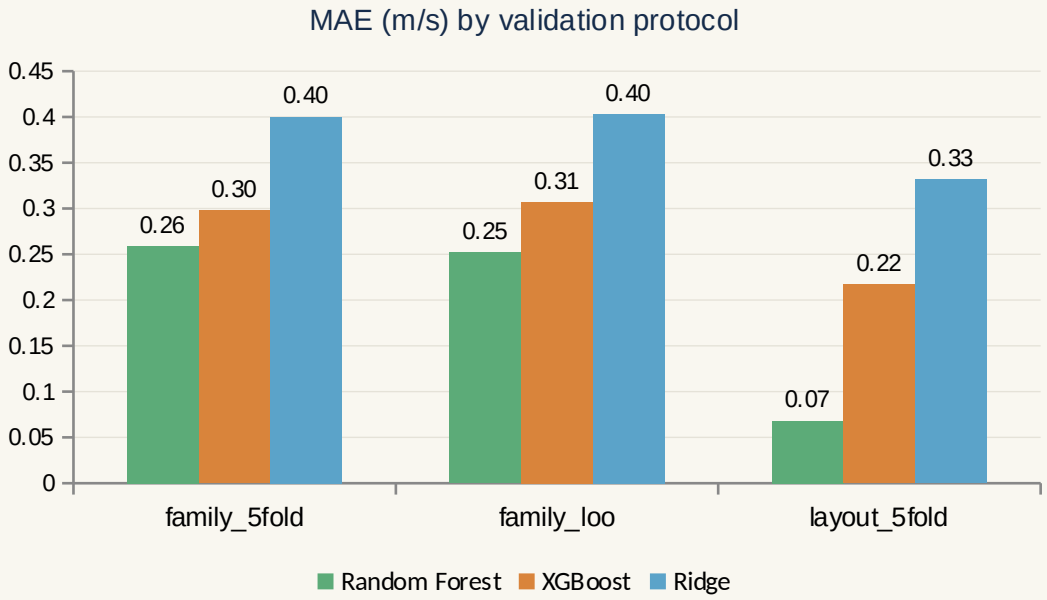
What the plot shows

Tree models follow the 1:1 line for most cases; the mean baseline is flat by construction.

Where it fails

Errors grow for observed speeds above ~4 m/s, which come from the sparsest layouts and are rare in the data.

Holding out sibling layouts instead of families makes Random Forest look about 4× better



The optimistic number

Random Forest MAE falls from 0.259 to 0.068 m/s when sibling layouts stay in training.

The honest number

Family-held-out MAE is the real generalisation claim; layout splits mostly measure interpolation.

Random Forest and XGBoost cannot be separated; engineered features only help the linear model

95% CI for MAE (family bootstrap)

Random Forest 0.18–0.34
XGBoost 0.24–0.36
Ridge 0.31–0.50
Mean baseline 0.57–1.08
Ridge and the baseline barely overlap with the trees.

Paired RF vs XGBoost

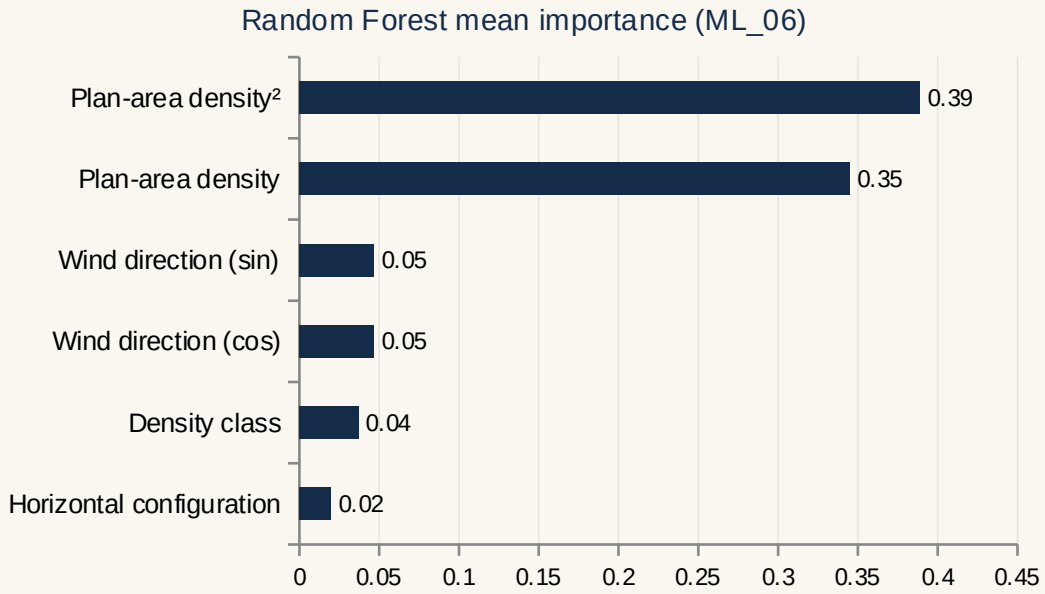
RF MAE is 0.039 m/s lower, but the 95% interval (−0.083 to +0.011) spans zero. No reliable winner between the two tree models.

Feature ablation

Engineered vs baseline features, MAE change: Ridge improves by 0.039 m/s; Random Forest worsens by 0.018; XGBoost by 0.002.

Fixed model settings and 16 families: treat rankings as indicative, not definitive.

Plan-area density and its square carry about 73% of Random Forest importance



Consistent with EDA

Density has the strongest raw link to wind speed (Pearson r = −0.75): denser canopies slow the flow.

Read with care

Importances are descriptive, not causal. Height features add little here; wind direction adds a small amount.

What this phase cannot claim

Scalar output

No wakes, street-canyon detail or velocity maps.

Small sample

224 cases, 112 layouts, 16 families.

Two directions

Only 0° and 45°; no claim for other angles.

Wind only

No temperature or heat prediction.

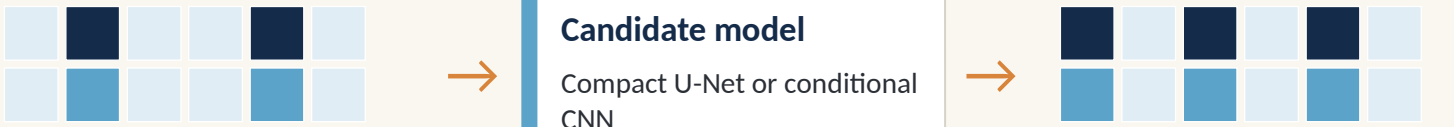
Fixed settings

No nested tuning; rankings may shift.

Idealised cases

Not a substitute for safety-critical design approval.

Phase 2: move from one number to a full velocity field



Evaluate on held-out layouts with field MAE, normalised RMSE and spatial error maps. Proposed direction, not an implemented result.

Key takeaway

Layout summaries predict canopy wind speed to ~0.26 m/s on unseen families, and honest family-level validation is what makes that claim credible.

APPENDIX

Team contributions and reproducibility

Team member	Student ID	Contribution
Suhaani Garg	2401010462	ML_01
Jashvita Lakshmi Omkaram	2401010310	ML_03–ML_04
Aarya Srivastava	2401010008	ML_02
Palak	2401010042	ML_05
Samarth Sangtani	2401010408	ML_06–ML_07

Run notebooks ML_01 → ML_07 in order. Repository: github.com/aarya-sr/T006-urban-microclimate-modelling

Note: ML_06 reports its own pooled Random Forest/XGBoost run (XGBoost MAE 0.232); all headline numbers here use the ML_07 comparison.